

Control Problems for Planar Maneuvers of a Spacecraft with Fuel Slosh

Mahmut Reyhanoglu¹, Sangbum Cho², and N. Harris McClamroch²

¹Department of Physical Sciences
Embry-Riddle Aeronautical University
Daytona Beach, FL 32114

²Department of Aerospace Engineering
University of Michigan
Ann Arbor, MI 48109

ABSTRACT: We study control problems for planar maneuvers of a spacecraft with fuel slosh in a zero gravity environment. The spacecraft is represented as a rigid body and fuel slosh dynamics are included using a single vibrating mass model. Both pendulum and spring-mass analogies are considered. The control inputs are defined by a thrust along the spacecraft's longitudinal axis, a transverse body fixed force and a pitching moment about the center of mass of the spacecraft; the slosh dynamics are assumed to be unactuated. The control objective is to design a feedback controller so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while suppressing the slosh mode. We formulate and analyze mathematical models that reflect this control objective.

KEYWORDS: Spacecraft, fuel slosh, feedback control

1. INTRODUCTION

It is well known that during thrust maneuvers the sloshing of fuel in partially filled tanks can interact with the controlled system in such a way as to cause the overall system to be unstable. Hence, the slosh dynamics should be included in the dynamic model of the spacecraft used for control synthesis. There is an extensive literature (see e.g.[1], [3], [8], [9]) on the interaction of vehicle dynamics and slosh dynamics and their control, but this literature treats only the case of small perturbations to the vehicle dynamics rather than the nonlinear translational and rotational vehicle dynamics that are treated in this paper.

With today's large and complex spacecraft, a substantial mass of fuel is necessary to place them into orbit and to perform orbital maneuvers. The mass of fuel contained in the tanks of a geosynchronous satellite amounts to approximately 40% of

its total mass [8]. When the fuel tanks are only partially filled, large quantities of fuel move inside the tanks under translational and rotational accelerations and generate the slosh dynamics. The slosh dynamics interacts with the rigid body dynamics of the spacecraft. Several methods have been employed to reduce the effect of sloshing, such as introducing baffles inside the tanks or dividing a large container into a number of smaller ones. These techniques, although helpful in some cases, do not completely succeed in cancelling the sloshing effects.

In this paper we consider a spacecraft with a spherical fuel tank and include the lowest frequency slosh mode in the dynamic model using pendulum and spring-mass analogies. We assume that a complete set of spacecraft control forces and moments is available to accomplish planar maneuvers. We ignore aerodynamic effects in our analysis, although they can be included in the spacecraft dynamics if we assume that they are cancelled by the spacecraft controls. We also assume zero gravity, but this assumption is for convenience only. These simplifying assumptions render the problem tractable, while still reflecting the important coupling between the unactuated slosh dynamics and the actuated rigid body motion of the spacecraft. Our control objective, as is typical for spacecraft maneuvering problems, is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the slosh mode. We subsequently construct and analyze mathematical models that reflect all of these assumptions and the control objective.

2. MODEL FORMULATION: PENDULUM ANALOGY

In this section, we formulate a mathematical model for a spacecraft with a spherical fuel tank and include the lowest frequency slosh mode in the dynamic model. This slosh mode is well approximated by considering the fuel to be “frozen” into a lump that slides on the sides of the tank. The fuel mass acts like a pendulum with its support point at the center of the tank as shown in Figure 1.

Consider a rigid spacecraft moving in a fixed plane as indicated in Figure 1. The important variables are the axial and transverse components of the velocity vector, v_x , v_z , the attitude angle θ of the spacecraft with respect to a fixed reference, and the angle ψ of the pendulum with respect to the spacecraft longitudinal axis, representing the fuel slosh. A thrust F , which is assumed to act through the spacecraft center of mass along the spacecraft’s longitudinal axis, a transverse force f , and a pitching moment M are available for control purposes. The constants in the problem are the spacecraft mass m and moment of inertia I (without fuel), the fuel mass m_f and moment of inertia I_f (assumed constant), the length $a > 0$ of the pendulum, and the distance b between the pendulum point of attachment and the spacecraft center of mass location along the longitudinal axis; if the pendulum point of attachment is in front of the spacecraft center of mass then $b > 0$. The parameters m_f , I_f and a

depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank.

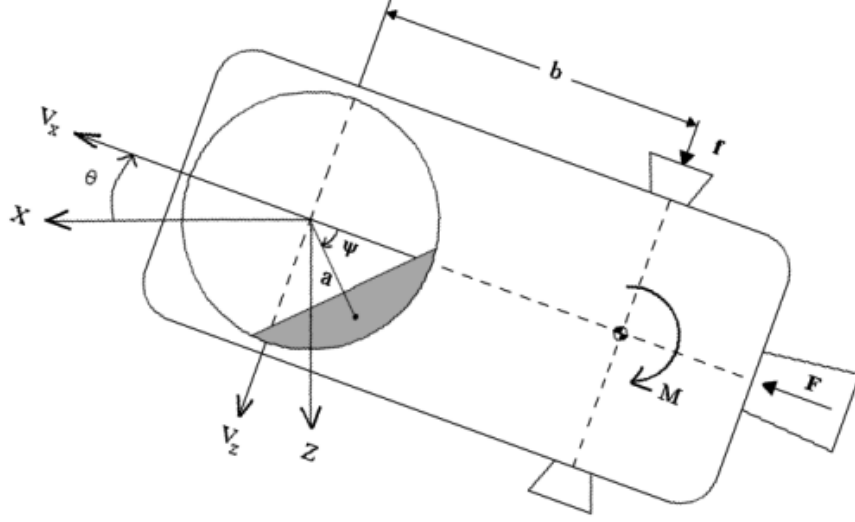


Figure 1: Model of a Spacecraft with Slosh Dynamics: Pendulum Analogy.

Under the indicated assumptions, the total kinetic energy is given by

$$T = \frac{1}{2}m[(v_x + b\dot{\theta} \sin \theta)^2 + (v_z + b\dot{\theta} \cos \theta)^2] + \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}I_f(\dot{\theta} + \dot{\psi})^2 \\ + \frac{1}{2}m_f[(v_x + a(\dot{\theta} + \dot{\psi}) \sin(\theta + \psi))^2 + (v_z + a(\dot{\theta} + \dot{\psi}) \cos(\theta + \psi))^2] .$$

Since gravitational effects are ignored, there is no potential energy. The resulting equations of motion are given by

$$(m + m_f)(\dot{v}_x + \dot{\theta}v_z) + m_fa(\ddot{\theta} + \ddot{\psi}) \sin \psi + mb\ddot{\theta}^2 + m_fa(\dot{\theta} + \dot{\psi})^2 \cos \psi = F , \\ (m + m_f)(\dot{v}_z - \dot{\theta}v_x) + m_fa(\ddot{\theta} + \ddot{\psi}) \cos \psi + mb\ddot{\theta} - m_fa(\dot{\theta} + \dot{\psi})^2 \sin \psi = f , \\ (I + mb^2)\ddot{\theta} + mb(\dot{v}_z - \dot{\theta}v_x) = M + bf , \\ (I_f + m_fa^2)(\ddot{\theta} + \ddot{\psi}) + m_fa[(\dot{v}_x + \dot{\theta}v_z) \sin \psi + (\dot{v}_z - \dot{\theta}v_x) \cos \psi] = 0 .$$

Define new velocity variables

$$v_1 = v_x \cos \psi - v_z \sin \psi ,$$

$$v_2 = v_x \sin \psi + v_z \cos \psi .$$

By introducing control transformations from (F, f, M) to new control acceleration inputs (u_1, u_2, u_3) the equations of motion can be written in $(v_1, v_2, \theta, \dot{\theta}, \psi, \dot{\psi})$ variables as

$$\dot{v}_1 = u_1 , \quad (1)$$

$$\dot{v}_2 = u_2 , \quad (2)$$

$$\ddot{\theta} = u_3 , \quad (3)$$

$$\ddot{\psi} = -cu_2 - u_3 + c(\dot{\theta} + \dot{\psi})v_1 , \quad (4)$$

where $c = m_f a / (I_f + m_f a^2)$. Equations (1)-(4) define a nonlinear control system with control $u = (u_1, u_2, u_3)^T$ and state $q = (v_1, v_2, \theta, \dot{\theta}, \psi, \dot{\psi})^T$.

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while attenuating the slosh mode.

Consider the nonlinear control system (1)-(4). The desired equilibrium solution for $u = (0, 0, 0)^T$ is given by $q^* = (V, 0, \theta^*, 0, 0, 0)^T$, i.e. we assume that the desired flight conditions are given by the translational velocity vector $(v_x^*, v_z^*) = (V, 0)$ and the attitude θ^* . Note that this equilibrium solution corresponds to the motion of the spacecraft with inertial velocity V (in the longitudinal direction) and attitude θ^* while the pendulum representing the fuel lump is aligned with the spacecraft's longitudinal axis.

Clearly, the linearized control system is not stabilizable. Moreover, it can be shown that there does not exist any time-invariant continuous feedback controller which asymptotically stabilizes the equilibrium but there do exist both time-invariant piecewise analytic feedback laws and time-periodic continuous feedback laws that asymptotically stabilize the equilibrium. The reader is referred to [4], [5] for a detailed controllability and stabilizability analysis applicable to a class of underactuated mechanical systems including the system considered in this paper. In our previous work [2], we considered a constant thrust $T > 0$ and designed a feedback law (using a Lyapunov approach) to stabilize the system to a relative equilibrium defined by a constant acceleration in the axial direction.

3. MODEL FORMULATION: SPRING-MASS ANALOGY

In this section, we formulate the dynamics of a spacecraft with a spherical fuel tank and include the lowest frequency slosh mode in the dynamic model using spring-mass analogy as shown in Figure 2. The spring-mass analogy is related to the pendulum analogy, in which the oscillation frequency of the mass-spring element represents the

lowest frequency sloshing mode [8].

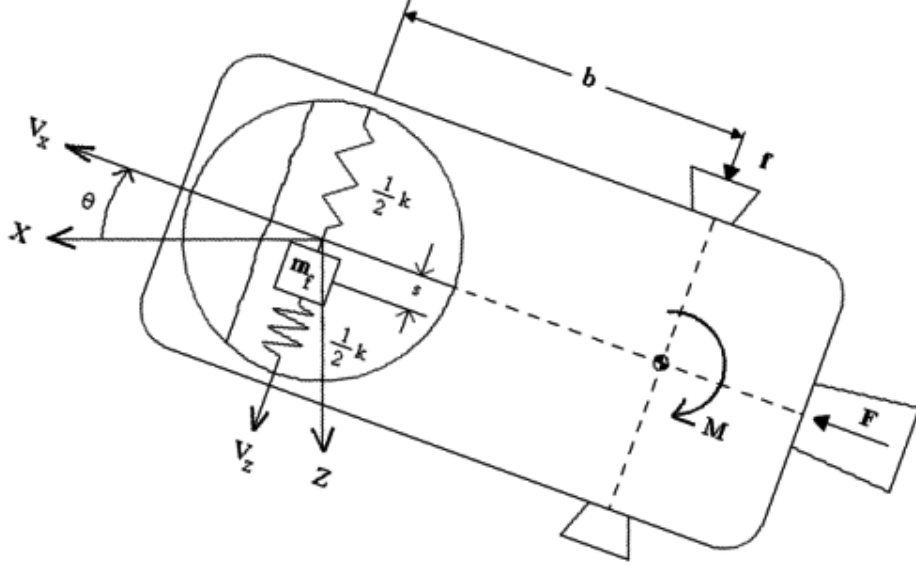


Figure 2: Model of a Spacecraft with Slosh Dynamics: Spring-Mass Analogy.

Consider a rigid spacecraft moving on a plane as indicated in Figure 2, where v_x , v_z are the axial and transverse components of the velocity vector and θ denotes the attitude angle of the spacecraft with respect to a fixed reference. The slosh mode is modeled by a point mass m_f whose relative position along the body z -axis is denoted by s ; a restoring force $-ks$ acts on the mass whenever the mass is displaced from its neutral position $s = 0$. As in the previous model, a thrust F , which is assumed to act through the spacecraft center of mass along the spacecraft's longitudinal axis, a transverse force f , and a pitching moment M are available for control purposes. The constants in the problem are the spacecraft mass m and moment of inertia I , the fuel mass m_f , and the distance b between the body z -axis and the spacecraft center of mass location along the longitudinal axis. The parameters m_f , k and b depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank.

Under the indicated assumptions, the Lagrangian is given by

$$L = \frac{1}{2}m[(v_x^2 + (v_z + b\dot{\theta})^2] + \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}m_f[(v_x + s\dot{\theta})^2 + (v_z + \dot{s})^2] - \frac{1}{2}ks^2 .$$

The resulting equations of motion are given by

$$\begin{aligned}
(m + m_f)(\dot{v}_x + \dot{\theta}v_z) + m_fs\ddot{\theta} + mb\dot{\theta}^2 + 2m_f\dot{s}\dot{\theta} &= F , \\
(m + m_f)(\dot{v}_z - \dot{\theta}v_x) + mb\ddot{\theta} + m_f\ddot{s} - m_fs\dot{\theta}^2 &= f , \\
(I + mb^2 + m_fs^2)\ddot{\theta} + m_fs(\dot{v}_x + \dot{\theta}v_z) + mb(\dot{v}_z - \dot{\theta}v_x) + 2m_fs\dot{s}\dot{\theta} &= M + bf , \\
m_f(\ddot{s} + \dot{v}_z - \dot{\theta}v_x - s\dot{\theta}^2) + ks &= 0 .
\end{aligned}$$

By introducing control transformations from (F, f, M) to new control acceleration inputs (u_1, u_2, u_3) the equations of motion can be written in $(v_x, v_z, \theta, \dot{\theta}, s, \dot{s})$ variables as

$$\dot{v}_x = u_1 , \quad (5)$$

$$\dot{v}_z = u_2 , \quad (6)$$

$$\ddot{\theta} = u_3 , \quad (7)$$

$$\ddot{s} = -u_2 + \dot{\theta}v_x + s\dot{\theta}^2 - k's , \quad (8)$$

where $k' = k/m$. Equations (5)-(8) define a nonlinear control system with control $u = (u_1, u_2, u_3)^T$ and state $q = (v_x, v_z, \theta, \dot{\theta}, s, \dot{s})^T$.

The control objective is again to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while attenuating the slosh mode.

Consider the nonlinear control system (5)-(8). The desired equilibrium solution for $u = (0, 0, 0)^T$ is given by $q^* = (V, 0, \theta^*, 0, 0, 0)^T$, i.e. we assume that the desired flight conditions are given by the translational velocity vector $(v_x^*, v_z^*) = (V, 0)$ and the attitude θ^* . Note that this equilibrium solution corresponds to the motion of the spacecraft with inertial velocity V (in the longitudinal direction) and attitude θ^* .

It is easy to show that if $k \neq 0$, the linearized control system is completely controllable and, hence, a linear feedback law can be designed to locally asymptotically stabilize the system (5)-(8). It is also possible to design discontinuous feedback laws that provide large guaranteed domain of attractions consistent with large angle flight maneuvers. However, if $k = 0$, then the system is not linearly stabilizable; in this case, discontinuous feedback controllers can be designed to achieve the control objective as demonstrated in our previous work ([6], [7]).

4. CONCLUSIONS

In this paper, we have considered single-mass models representing the lowest frequency slosh mode. There are a number of interesting extensions and modifications that could be made to the model and control problems defined in this paper. These include incorporation of multi-mode slosh dynamics, incorporation of gravitational effects, and development of full three dimensional models.

ACKNOWLEDGEMENT

M. Reyhanoglu wishes to acknowledge the support provided by Embry-Riddle Aeronautical University. N.H. McClamroch acknowledges the partial support provided by NSF Grant ECS-9625173.

REFERENCES

- [1] Bryson, A.E., Control of Spacecraft and Aircraft, Princeton University Press, Princeton, 1982.
- [2] Cho, S., N. H. McClamroch, and M. Reyhanoglu, "Feedback Control of a Space Vehicle with Unactuated Fuel Slosh Dynamics," to appear in AIAA Guidance, Navigation and Control Conference, Denver, CO, August 2000.
- [3] Peterson, L.D., E.F. Crawley and R.J. Hansman, "Nonlinear Fluid Slosh Coupled to the Dynamics of a Spacecraft," AIAA Journal, **27**(9), 1989, pp.1230-1240.
- [4] Reyhanoglu, M., A.J. van der Schaft, N.H. McClamroch and I. Kolmanovsky, "Nonlinear Control of a Class of Underactuated Systems," Proceedings of IEEE Conference on Decision and Control, 1996, Kobe, Japan, pp.1682-1687.
- [5] Reyhanoglu, M., A.J. van der Schaft, N.H. McClamroch and I. Kolmanovsky, "Dynamics and Control of a Class of Underactuated Mechanical Systems," IEEE Transactions on Automatic Control, **44**(9), September 1999, pp.1663-1671.
- [6] Reyhanoglu, M., S. Cho, N.H. McClamroch, "Feedback Control for Planar Maneuvers of an Aerospace Vehicle with an Underactuated Internal Degree of Freedom," Proceedings of American Control Conference, 1999, San Diego, CA, pp.3432-3436.
- [7] Reyhanoglu, M., S. Cho, N.H. McClamroch, "Discontinuous Feedback Control of a Special Class of Underactuated Mechanical Systems," International Journal of Robust and Nonlinear Control, **10**(4), 2000, pp.265-281.

[8] Sidi, M.J., Spacecraft Dynamics and Control, Cambridge University Press, Cambridge, 1997.

[9] Wie, B., Space Vehicle Dynamics and Control, AIAA Education Series, Reston, VA, 1998.